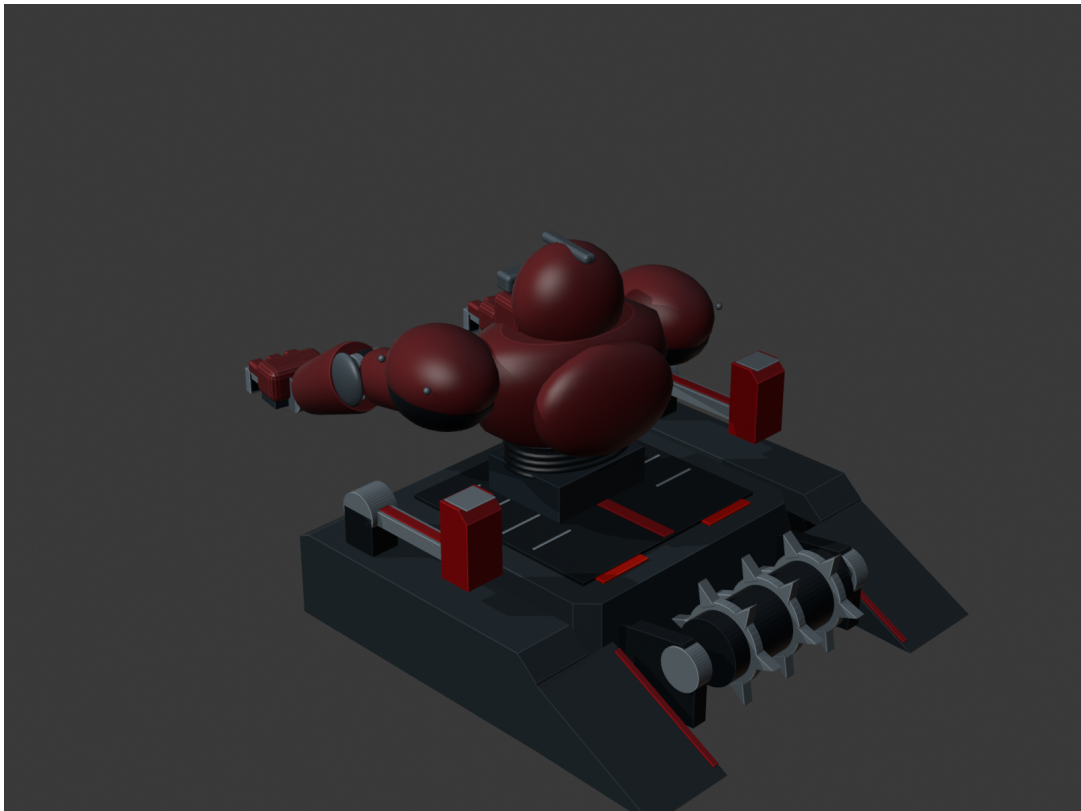


SENTINELT

Robo Wars RC Combat Robot

Holistic Build Document

Project Developer: TG Assegaai



Verified digital envelope: 318.16 mm x 495.00 mm x 405.99 mm | Footprint audit: 180/180 frames PASS

1. Context and Robo Wars Design Constraints

SentinelT is a human-operated 2.4 GHz RC combat robot. The design addresses the Robo Wars constraints: maximum footprint 500 mm x 500 mm, mass strictly below 5 kg, an accessible ON/OFF switch, a mandatory Emergency Stop power-cut mechanism, 2.4 GHz remote control, and no projectiles, flames or liquids.

Constraint	SentinelT response
Footprint	318.16 mm x 495.00 mm; 180/180 frames PASS
Mass	4.650 kg engineering allocation; 0.350 kg design margin
Control	Human-operated 2.4 GHz RC
Main power	Accessible ON/OFF battery isolation
E-Stop	Mushroom E-Stop controlling a separate high-current isolation stage
Restricted weapon types	No projectiles, flames or liquids

2. Solution Overview

I am building SentinelT as a compact armored Robo Wars robot with protected drive components, sloped external protection, a serviceable electronics bay, an articulated upper structure, a guarded active-mechanism provision, and a fail-safe power architecture. The design combines a 3S battery system, brushed drive control, 2.4 GHz RC input, safe-state logic, main battery isolation and an E-Stop-controlled high-current power cut.

3. Mechanical Design

The chassis uses a 6 mm HDPE structural/armor material plan, with metal hardware where shafts, motor mounts, brackets, terminals and fasteners require greater strength. Bolted construction supports repair, inspection and component replacement.

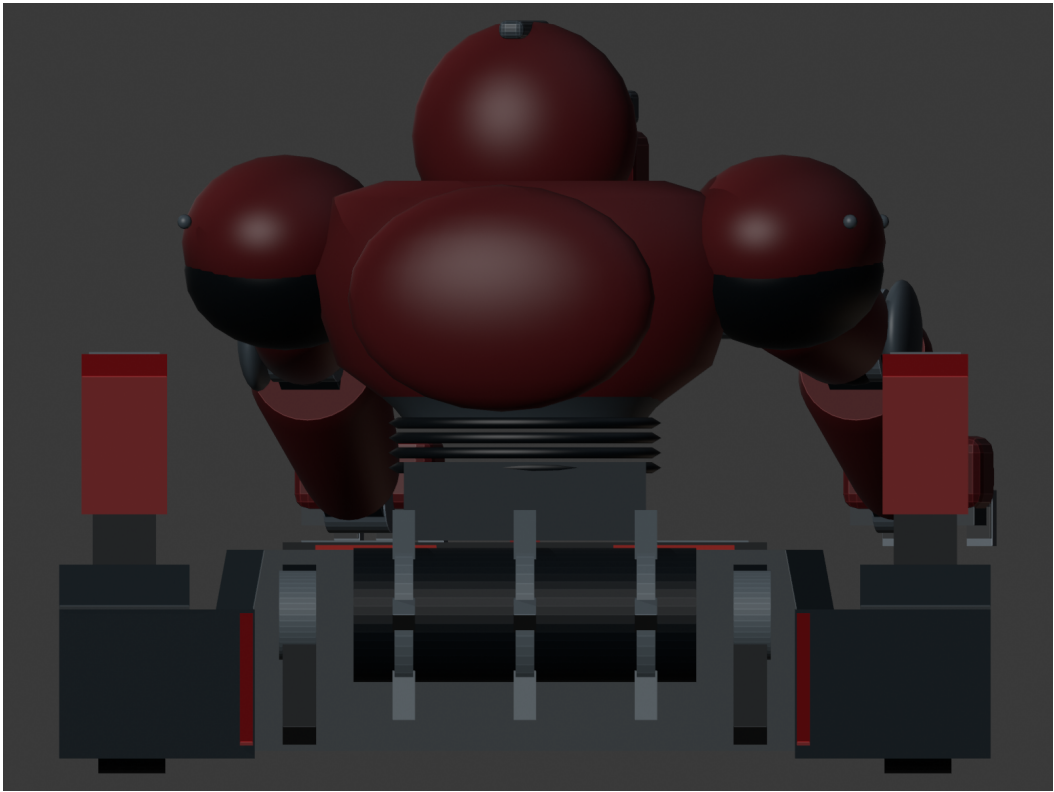


Figure 1. SentinelT front CAD view.

Mechanical feature	Design intent
HDPE chassis/armor	Impact-resistant, machinable and serviceable structure
Protected drive zones	Reduce direct exposure of motors and wheels
Sloped guards	Deflect contact away from flat vulnerable faces
Bolted joints	Support repair and internal access
Metal hardware	Carry concentrated drivetrain and fastening loads

4. Internal Packaging and Buildability

The service bay reserves space for the 3S battery, drive ESC, X6B receiver, fuse, power distribution, E-Stop relay/contactor, XT60 connector, main ON/OFF isolator, auxiliary controller and left/right drive motors.

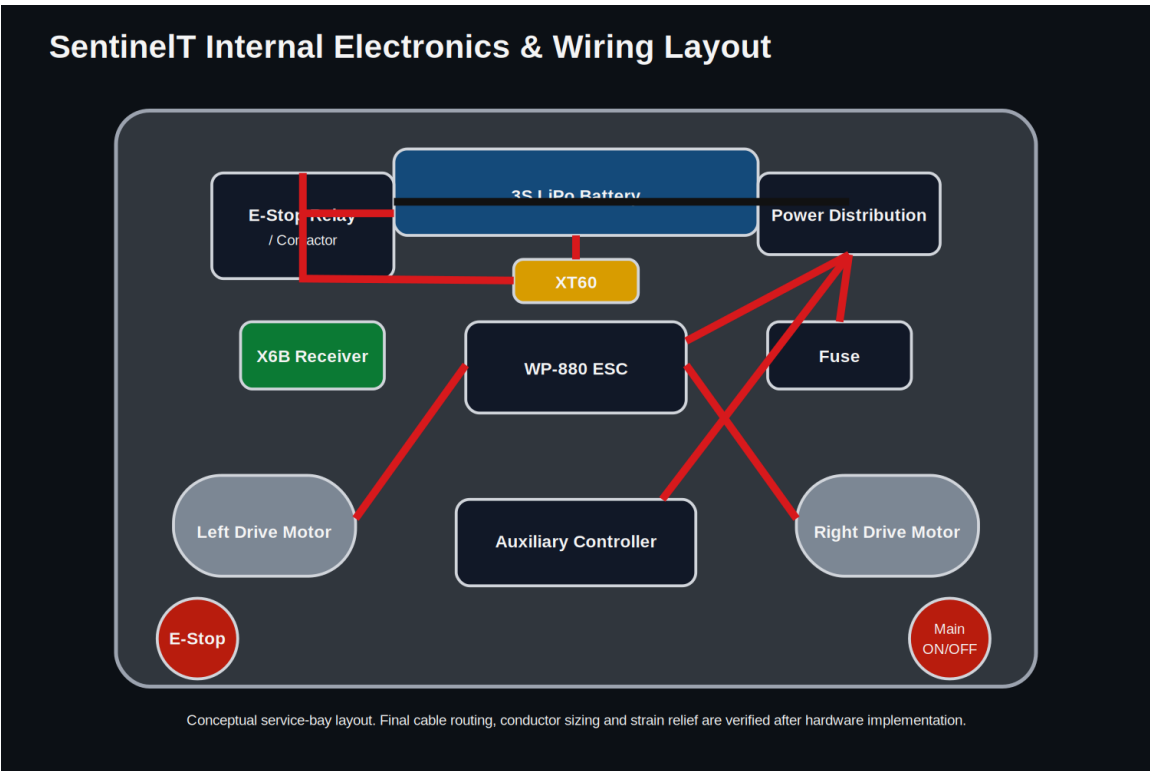


Figure 2. Internal electronics and wiring layout.

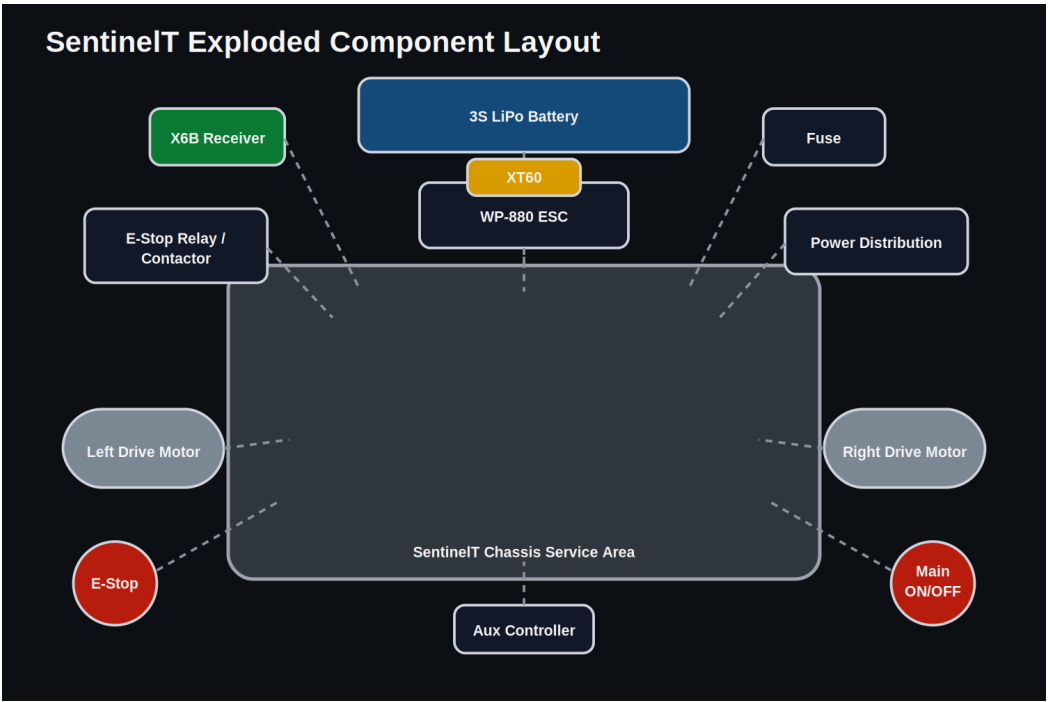


Figure 3. Exploded component layout.

5. Electronic Design

The electronic architecture uses a 3S LiPo source, high-current protection and isolation, a 2.4 GHz RC receiver, brushed drive control and a separate auxiliary branch. The main ON/OFF isolator and E-Stop are hardware safety functions rather than software-only controls.

Subsystem	Selected / planned hardware
Battery	3S 11.1 V 5000 mAh LiPo, XT60
RC system	FlySky FS-i6X transmitter + X6B receiver, 2.4 GHz
Drive controller	HobbyWing QuicRun WP-880 dual brushed ESC
Drive motors	2 x 12 V 37D 50:1 geared motors
Main isolation	12 V high-current battery isolator
E-Stop	Mushroom pushbutton controlling a high-current relay/contactor
Distribution	High-current distribution block and fused branches
Auxiliary branch	Guarded active-mechanism controller/actuator; final selection after hardware matching

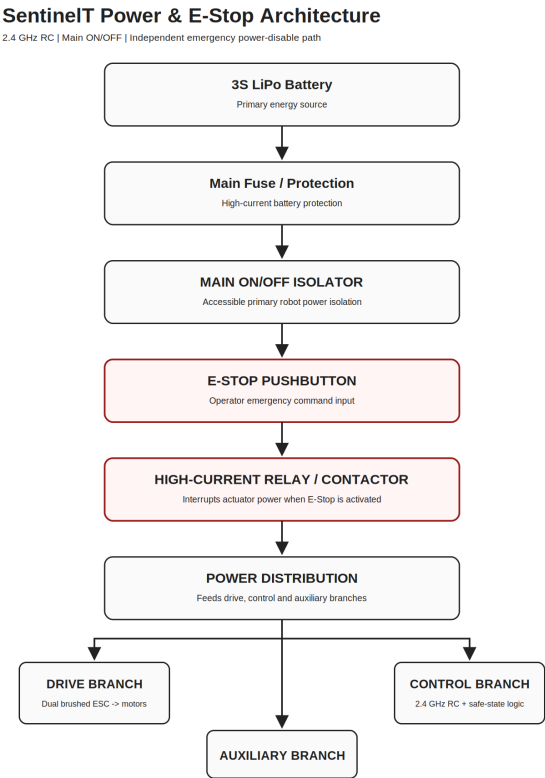


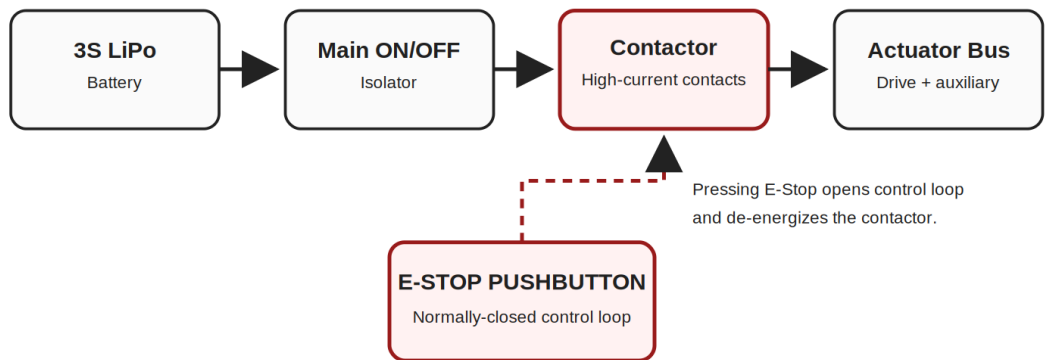
Figure 4. SentinelT power and E-Stop architecture.

6. E-Stop Integration and Wiring

The Emergency Stop is designed as an independent power-disable path. The mushroom switch opens the control circuit of a suitably rated relay/contactors; the contactor main contacts then interrupt actuator power downstream of the main isolator. Software safe-state behavior is an additional layer and is not the only means of stopping powered actuators.

SentinelT E-Stop Integration

Fail-safe concept: the mushroom switch controls a separate high-current isolation device.



The E-Stop does not rely on software to remove actuator power. Software safe-state logic is an additional control layer.

Figure 5. Mandatory E-Stop integration.

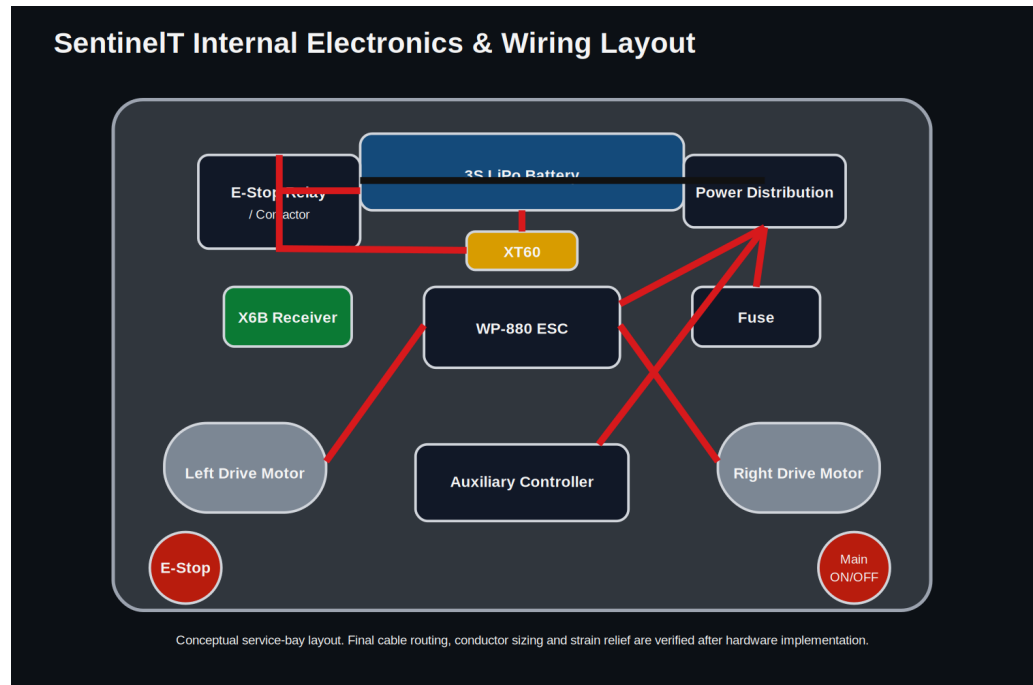


Figure 6. Internal wiring/component placement evidence.

7. Programming and Framework Design

The control framework uses C/C++ with Arduino framework conventions and a deterministic state model: `SAFE_DISABLED`, `RC_READY` and `OPERATING`. Human RC input is checked for validity before motion is allowed. Differential drive mixing uses throttle and steering commands. Startup, reset, invalid command data, RC signal loss and an active E-Stop force the software to the disabled state.

Programming element	Implementation
Language / framework	C/C++ using Arduino framework conventions
Input method	2.4 GHz RC receiver channel data
Drive method	Differential mixing: left = throttle + steering; right = throttle - steering
RC safety timeout	250 ms validity timeout
Default state	All actuator commands disabled
Auxiliary logic	Explicit operator enable; default OFF
Hardware parameters	Receiver protocol, pin mapping and ESC calibration verified after hardware implementation

SentinelT 2.4 GHz RC Control Flow

Safe-state logic for human-operated Robo Wars control

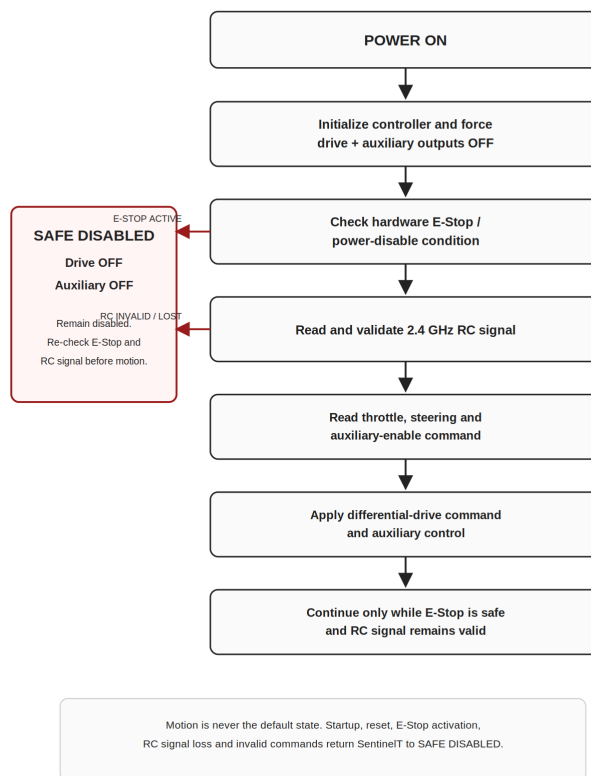


Figure 7. SentinelT RC control and safe-state flow.

8. Simulation and Digital Verification

Blender and Python-based dimensional auditing are used before physical fabrication. The current audit evaluates the world-space bounds of 204 physical mesh objects over frames 1-180 and compares the X and Y extents with the 500 mm x 500 mm limit.

Audit result	Value
Physical mesh objects	204
Maximum X	318.16 mm at frame 1
Maximum Y	495.00 mm at frame 1
Maximum Z	405.99 mm
Frames checked	180
Failed frames	0
Status	PASS - 180/180 frames

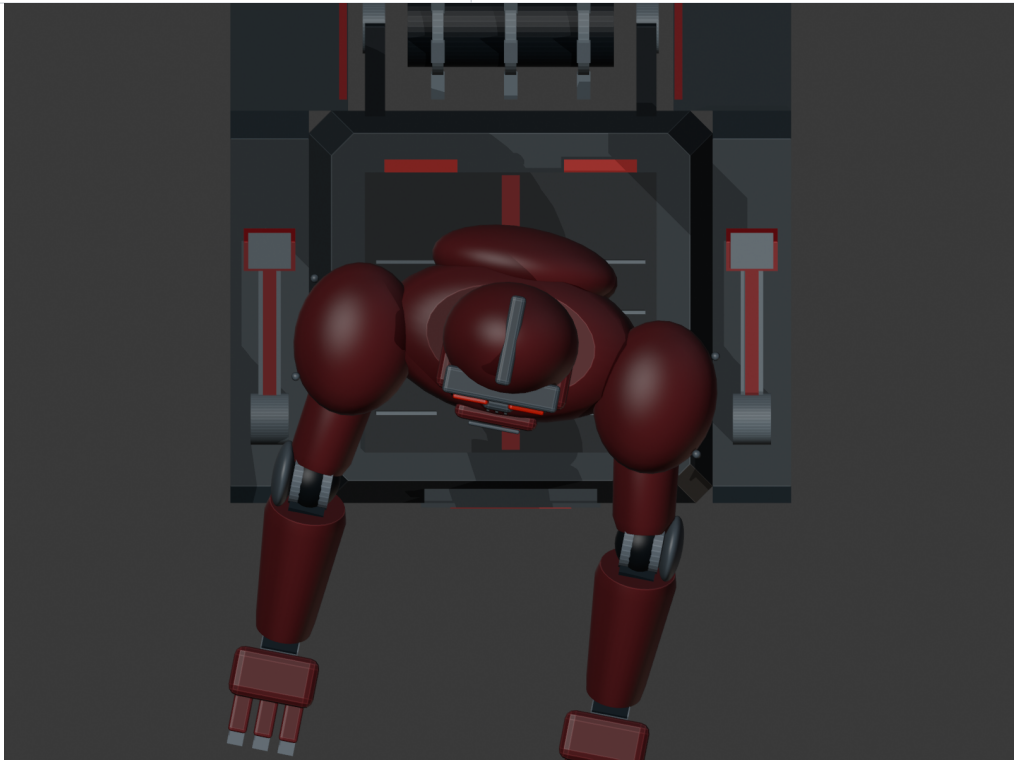


Figure 8. Top CAD view supporting footprint verification.

9. Engineering Mass Budget

The 4.650 kg value is an engineering allocation, not a measured physical mass. It leaves a 0.350 kg design margin below the 5.000 kg limit. Physical weighing will be completed after hardware implementation.

Subsystem	Allocated mass
Chassis and lower armor	1.10 kg
Drive motors and gearboxes	0.75 kg
Wheels, hubs and shafts	0.30 kg
Battery and power system	0.60 kg
RC electronics, ESC and wiring	0.25 kg
Guarded active mechanism + mount	0.70 kg
Upper structure and outer shells	0.50 kg
Bearings, brackets and fasteners	0.25 kg
Contingency	0.20 kg
TOTAL	4.650 kg

10. BOM Summary

The full Bill of Materials is stored in Documentation/BOM/SentinelT_Bill_of_Materials.xlsx. It records component name, quantity, cost per unit, supplier, supplier stock code, direct purchase URL and line/total cost information. The current priced component subtotal is R8,039.15. The guarded active-mechanism motor/controller remains to be finalized after hardware matching.

11. Current Implementation Status

The digital engineering package contains CAD geometry, footprint verification, internal component layout, electronic architecture, E-Stop integration, control framework, flowchart, BOM structure, pitch deck and holistic report. After hardware implementation, the remaining physical evidence will verify measured mass, current draw, thermal behavior, battery runtime, drivetrain response, cable routing and E-Stop interruption performance.

12. Project Development Responsibility

SentinelT is developed by TG Assegaa. Development responsibilities cover the system concept, Blender CAD modelling, mechanical packaging, dimensional verification scripts, 2.4 GHz control architecture, safe-state framework, power and E-Stop design, component research, Bill of Materials and technical documentation.

Project Developer: TG Assegaa

Signature: Thato Glen Assegaa